

Mark Scheme — Practice Paper 10 (P1)

M = method mark, A = accuracy mark, B = independent mark. 教师/批改用。

1 $x = 5 - 2y$: $(5 - 2y)^2 + y^2 = 10 \Rightarrow 5y^2 - 20y + 15 = 0 \Rightarrow y^2 - 4y + 3 = 0$ (M1 A1). $y = 1, 3$: $(3, 1)$ and $(-1, 3)$ (A1). [3]

2(a) $\binom{5}{3}(3x)^3 = 10 \times 27$ (M1) = 270 (A1). [2]

2(b) x^2 coefficient of $(1 + 3x)^5$ is $\binom{5}{2} \times 9 = 90$. x^3 coefficient = $2(270) - 1(90)$ (M1) = 450 (A1). [2]

3(a) $18 + 9\theta = 30$ (M1) $\Rightarrow \theta = \frac{4}{3}$ rad (A1). [2]

3(b) $\frac{1}{2}(9)^2 \left(\frac{4}{3}\right)$ (M1) = 54 cm² (A1). [2]

4(a) $a + 3d = 3a \Rightarrow 3d = 2a$ (B1). $3(2a + 5d) = 96 \Rightarrow 2a + 5d = 32$ (B1). Solve: $a = 6, d = 4$ (A1). [3]

4(b) $6 + 4(n - 1) = 106$ (M1) $\Rightarrow n = 26$ (A1). [2]

5(a) $f(x) < 4$ (B2; B1 for $f(x) \leq 4$ or $y < 4$ stated as " $x < 4$ "). [2]

5(b) $y = 4 - \frac{3}{x} \Rightarrow \frac{3}{x} = 4 - y \Rightarrow x = \frac{3}{4 - y}$ (M1). $f^{-1}(x) = \frac{3}{4 - x}$ (A1). Domain $x < 4$ (B1). [3]

5(c) $4 - \frac{3}{2x} = 1 \Rightarrow 2x = 1 \Rightarrow x = \frac{1}{2}$ (B1). [1]

6(a) $A(10, 0), B(0, 5)$ (B1 B1). [2]

6(b) Gradient $AB = -\frac{1}{2}$, so OC has gradient 2: $y = 2x$ (M1). $x + 2(2x) = 10 \Rightarrow x = 2, C(2, 4)$ (M1 A1). $OC = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$ (A1). [4]

7(a) $3 \sin \theta = 2(1 - \sin^2 \theta) \Rightarrow 2 \sin^2 \theta + 3 \sin \theta - 2 = 0$ (M1). $(2 \sin \theta - 1)(\sin \theta + 2) = 0 \Rightarrow \sin \theta = \frac{1}{2}$ (A1). $\theta = 30^\circ, 150^\circ$ (A1). [3]

7(b) $2x + 40^\circ$ lies in $[40^\circ, 400^\circ]$ (B1). $\sin = -0.5$ at $210^\circ, 330^\circ$ (M1). $2x = 170^\circ, 290^\circ \Rightarrow x = 85^\circ$ (A1), 145° (A1). [4]

8(a) $\frac{dy}{dx} = 6x^2 - 18x + 12 = 6(x - 1)(x - 2)$ (B1). Stationary at $x = 1, 2$: $(1, 2)$ and $(2, 1)$ (M1 A1).

$\frac{d^2y}{dx^2} = 12x - 18$: -6 at $x = 1$ (maximum), $+6$ at $x = 2$ (minimum) (A1). [4]

8(b) $6(x - 1)(x - 2) > 0$ (M1) $\Rightarrow x < 1$ or $x > 2$ (A1). [2]

8(c) At $x = 0$: $y = -3$, gradient 12, normal gradient $-\frac{1}{12}$ (M1). $y + 3 = -\frac{1}{12}x \Rightarrow x + 12y + 36 = 0$ (A1). [2]

9(a) $y = \int (2 - 8x^{-3}) dx = 2x + \frac{8x^{-2}}{2} + c = 2x + \frac{4}{x^2} + c$ (M1 A1). $(2, 5)$: $4 + 1 + c = 5 \Rightarrow c = 0$ (M1). $y = 2x + \frac{4}{x^2}$ (A1). [4]

9(b) Roots $x = 1, 3$ (B1). $\int_1^3 (x^2 - 4x + 3) dx = \left[\frac{1}{3}x^3 - 2x^2 + 3x\right]_1^3 = 0 - \frac{4}{3} = -\frac{4}{3}$ (M1). Area = $\frac{4}{3}$ (A1). [3]

9(c) $\left[-\frac{1}{2(2x + 1)}\right]_0^2$ (M1) = $-\frac{1}{10} + \frac{1}{2} = \frac{2}{5}$ (A1). [2]

$$\mathbf{10(a)} \quad \frac{(1 - \cos \theta) + (1 + \cos \theta)}{(1 + \cos \theta)(1 - \cos \theta)} \text{ (M1)} = \frac{2}{1 - \cos^2 \theta} \text{ (M1)} = \frac{2}{\sin^2 \theta} \text{ (A1 AG). [3]}$$

$$\mathbf{10(b)} \quad \sin^2 \theta = \frac{1}{4} \text{ (M1)} \Rightarrow \sin \theta = \frac{1}{2} \text{ (positive in } 0^\circ < \theta < 180^\circ \text{) (A1). } \theta = 30^\circ, 150^\circ \text{ (A1). [3]}$$

$$\mathbf{10(c)} \quad \cos x = -\sqrt{1 - \frac{4}{9}} = -\frac{\sqrt{5}}{3} \text{ (M1; negative since } x \text{ obtuse). } \tan x = \frac{2/3}{-\sqrt{5}/3} = -\frac{2}{\sqrt{5}} = -\frac{2\sqrt{5}}{5} \text{ (A1). [2]}$$

$$\mathbf{11(a)} \quad ar = 6, \frac{a}{1-r} = 27 \Rightarrow 27r(1-r) = 6 \text{ (M1)} \Rightarrow 9r^2 - 9r + 2 = 0 \Rightarrow (3r-1)(3r-2) = 0$$

$$\text{(A1). } r = \frac{1}{3}, a = 18 \text{ (A1); } r = \frac{2}{3}, a = 9 \text{ (A1). [4]}$$

$$\mathbf{11(b)(i)} \quad u_n = S_n - S_{n-1} = (2n^2 + 3n) - (2(n-1)^2 + 3(n-1)) \text{ (M1)} = 4n + 1 \text{ (A1). } u_{n+1} - u_n = 4, \text{ constant, so arithmetic with } a = 5, d = 4 \text{ (A1). [3]}$$

$$\mathbf{11(b)(ii)} \quad S_{20} - S_{10} = (800 + 60) - (200 + 30) \text{ (M1)} = 630 \text{ (A1). [2]}$$

$$\mathbf{12(a)} \quad (x+2)^2 + (y-1)^2 = 25: \text{ centre } (-2, 1) \text{ (B1), radius } 5 \text{ (B1). [2]}$$

$$\mathbf{12(b)} \quad 1 + 25 + 4 - 10 - 20 = 0 \checkmark \text{ (B1). Radius gradient } \frac{5-1}{1+2} = \frac{4}{3} \text{ (M1), tangent gradient } -\frac{3}{4} \text{ (A1).}$$

$$y - 5 = -\frac{3}{4}(x - 1) \Rightarrow 3x + 4y = 23 \text{ (A1). [4]}$$

Total: 75